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1. Introduction

The National Health and Nutrition Examination Survey (NHANES), conducted by the Centers for Disease Control and Prevention (CDC), is one of the most widely used sources of health data in the United States. Unlike many observational datasets, NHANES is based on a complex, multistage probability sampling design that allows researchers to make valid inferences about the civilian, noninstitutionalized U.S. population. By combining interview responses, physical examinations, and laboratory measurements, NHANES provides a uniquely rich framework for studying relationships between health outcomes, behavioral factors, and demographic characteristics.

However, the complexity of the NHANES sampling design introduces important statistical considerations. The use of stratification, clustering, and unequal sampling weights means that standard methods assuming simple random sampling are not appropriate. Analyses that ignore the design can lead to biased estimates of variability and overly confident conclusions. As a result, valid inference requires a design-based approach that explicitly incorporates survey weights, primary sampling units, and strata.

In this report, we analyze data from the 2015–2016 NHANES cycle to produce finite population inferences for key health-related quantities, including mean total cholesterol levels, subgroup

differences across demographic categories, and the relationship between cholesterol and covariates such as age, gender, and race/ethnicity. Using the survey package in R, we construct a design object that reflects the NHANES sampling structure and use it to obtain point estimates, standard errors, and confidence intervals that properly account for the survey design.

Beyond estimating population parameters, a central goal of this paper is to evaluate how the complex survey design affects statistical inference. In particular, we compare design-based estimates to naive estimates that ignore the sampling structure in order to illustrate the impact on variance and efficiency. We also examine subgroup variation and regression relationships to demonstrate how design-based methods extend beyond simple descriptive statistics.

Overall, this analysis highlights both the substantive insights that can be drawn from NHANES data and the methodological importance of properly accounting for complex survey designs. By integrating applied statistical analysis with a discussion of survey methodology, this report provides a clear example of how design-based inference supports accurate and meaningful conclusions in real-world population studies.

2. Data

For the purposes of this study, our group analyzed data from the 2015 - 2016 cycle of the National Health and Nutrition Examination Survey (NHANES), a complex multistage probability survey designed to produce nationally representative estimates of health and nutrition for the civilian, noninstitutionalized U.S. population.

2.1 Source

The data used in this analysis come from the publicly available 2015–2016 cycle of the National Health and Nutrition Examination Survey (NHANES), administered by the Centers for Disease

Control and Prevention (CDC). NHANES is designed to produce nationally representative estimates for the civilian, noninstitutionalized U.S. population through a complex, multistage probability sampling framework that incorporates stratification, clustering, and unequal probabilities of selection.

For this study, we used demographic data, laboratory measurements of total cholesterol, and key covariates including age, gender, and race/ethnicity. These data originate from separate NHANES files and were combined at the individual level using the respondent sequence number ([SEQN](#)), which uniquely identifies each participant across datasets. The merging process produced a unified analytic dataset containing all variables required for estimation.

The implementation of the NHANES data structure and survey design followed guidance from the [asdfree](#) NHANES repository, which provides reproducible R code and documentation for constructing survey design objects and conducting design-based inference. After merging and restricting to relevant variables, the resulting dataset contained 7,256 individuals before weighting, representing a much larger target population once survey weights were applied.

2.2 Pre-processing

All datasets were imported into R and merged using [SEQN](#). Variables of interest, including total cholesterol, age, gender, and race/ethnicity, were selected and formatted for analysis. Categorical variables were converted into factor form to support subgroup estimation and regression modeling. The analytic dataset was constructed so that all subsequent calculations could be performed consistently within a single merged structure rather than across separate files.

Handling missing data required distinguishing between the NHANES survey methodology and our own analytic choices. NHANES incorporates adjustments for nonresponse within its weighting scheme and uses imputation for certain variables in the broader survey program.

However, the variables used in this analysis, including total cholesterol and the primary demographic covariates, were not drawn from multiple imputed NHANES datasets, and no additional imputation procedures were applied in this study.

Instead, we used a complete-case approach by excluding observations with missing values in key analysis variables. This ensured that all reported estimates were based on a consistent analytic sample across methods. While this approach simplifies implementation and aligns with the design-based workflow used throughout the analysis, it may introduce bias if the missingness is not random. As a result, the estimates should be interpreted as design-based inferences for the observed complete-case sample, with the understanding that NHANES survey weights partially adjust for nonresponse at the population level.

Because total cholesterol is measured in the Mobile Examination Center, the final analytic sample was carried forward into the survey design stage using the MEC examination weight (WTMEC2YR), along with the NHANES stratification (SDMVSTRA) and primary sampling unit (SDMVPSU) variables. This ensures that the complete-case sample still produces nationally representative estimates when analyzed within the survey design framework.

3. Sampling Design

NHANES uses a complex, multistage probability sampling design rather than a simple random sample. This distinction is central to the analysis because valid national inference depends not only on the values observed in the sample, but also on the way the sample was selected. In NHANES, respondents are chosen through a structured design that incorporates stratification, clustering, and unequal probabilities of selection. As a result, standard methods that assume independent observations drawn with equal probability are not appropriate for estimation or inference.

3.1 Multistage Design

NHANES selects participants through a four-stage sampling process. First, primary sampling units (PSUs), typically counties or groups of counties, are selected. Second, smaller geographic segments are sampled within those PSUs. Third, households are selected from within those segments. Finally, individuals are chosen from sampled households to complete interviews and clinical examinations. This design makes nationwide data collection practical while preserving broad population coverage, but it also means that the observed sample does not behave like a simple random sample of U.S. residents.

3.2 Stratification

Before the first stage of selection, geographic areas are organized into strata. In the computational design object, this structure is represented by the NHANES stratification variable `SDMVSTRA`. Stratification is used to improve precision and ensure that the sample reflects important geographic and demographic variation in the target population. By sampling independently within strata, NHANES can maintain national coverage while improving the stability of certain estimates relative to an unstructured design.

3.3 Clustering

NHANES also relies on clustering, which is represented in our analysis by the primary sampling unit variable `SDMVPSU`. Clustering reduces the logistical burden of fieldwork by concentrating data collection within selected geographic areas. However, individuals within the same cluster often share similar environmental, socioeconomic, or demographic characteristics, which creates positive intraclass correlation. This reduces the effective sample size and increases the variance of estimates compared with what would be expected under simple random sampling. For that reason, clustering must be incorporated directly into the design-based analysis rather than ignored.

3.4 Oversampling

NHANES intentionally oversamples certain subpopulations, including some minority groups and older adults, to improve the precision of subgroup estimates. Because of this, the raw sample is not proportionally representative of the national population on its own. Some groups appear more often in the observed data than they would under proportional allocation. This is why weighting is necessary: without it, unweighted estimates would reflect the design of the sample rather than the actual population composition.

3.5 Why the Weighted Mean Can Recover the Population Mean

An intuitive way to understand the weighted mean is to treat each sampled respondent as representing a different number of people in the national population. Respondents from oversampled groups receive smaller weights because each one stands in for fewer people, while respondents from less frequently selected groups receive larger weights because each one stands in for more people. The weighted mean, therefore, is not simply the average cholesterol level among the observed respondents. Instead, it is a design-based estimate of the national mean under the NHANES sampling plan. This is how a sample of 7,256 individuals can be used to estimate the mean cholesterol level for the entire civilian, noninstitutionalized U.S. population.

4. Sample Size and Weights

The multistage sampling design determines who enters the sample, while the weights determine how each sampled individual contributes to the national inference. For the 2015–2016 cycle, the analytic sample used in this study contained 7,256 respondents. Once the appropriate NHANES examination weights were applied, these observations represented an estimated civilian, noninstitutionalized U.S. population of approximately 271,045,833 people. This difference between the unweighted sample size and the weighted population size is what allows NHANES to support finite population inference from a comparatively small observed dataset.

4.1 Why WTMEC2YR Was Used

The weight used in this analysis was **WTMEC2YR**, the two-year Mobile Examination Center examination weight. This is the appropriate weight because total cholesterol is measured during the Mobile Examination Center component rather than in the interview-only component of NHANES. Choosing the correct weight is necessary because different NHANES variables correspond to different stages of participation and therefore to different respondent universes. In this case, **WTMEC2YR** ensures that the cholesterol estimates are properly scaled to the target population represented by examined respondents in the 2015–2016 cycle.

4.2 What the Weights Adjust For

The NHANES survey weights incorporate three main adjustments. First, they correct for unequal selection probabilities caused by oversampling. Second, they adjust for nonresponse so that respondents help represent individuals with similar characteristics who were selected but did not participate. Third, they are calibrated to external population benchmarks such as age, sex, and race/ethnicity totals, allowing the weighted sample to align with the known composition of the civilian, noninstitutionalized U.S. population. In this sense, the weights do not merely re-scale the sample; they connect the observed data to the broader population the survey is intended to represent.

4.3 Why Weighting Changes the Output

Weighting changes the output because the raw sample composition is shaped by the survey design rather than by population proportions alone. Unweighted estimates describe the observed respondents, but weighted estimates are intended to recover the corresponding national quantities. In our analysis, applying **WTMEC2YR** expanded the complete-case sample into a population estimate of about 271 million people. Weighting also affects subgroup precision, since groups

with smaller effective sample sizes or more variable weights tend to have larger standard errors. This explains why some demographic categories produce less precise estimates than others, even within the same survey cycle.

5. Estimation Framework

We use a framework that accounts for the NHANES sampling structure in order to draw reliable population-level inferences. This approach follows design-based inference, where population values are treated as fixed, and the sampling process is the primary source of randomness.

At the core of this framework is the Horvitz–Thompson estimator, which achieves unbiased estimation by weighting each observed unit by the inverse of its inclusion probability. From this foundation, all parameters of interest, including means, proportions, and regression coefficients, are derived. This ensures that estimates properly reflect the unequal selection probabilities embedded in the NHANES design.

Standard statistical functions such as **mean()** or **lm()** in R are not appropriate in this setting because they assume independent observations drawn with equal probability. These assumptions are violated in NHANES due to clustering, stratification, and unequal weighting. As a result, using these functions would produce biased standard errors and misleading inference, particularly by understating uncertainty.

To correctly implement the Horvitz–Thompson framework, we use the **survey** package in R. This package allows the analyst to construct a survey design object that incorporates the primary sampling units (**SDMVPSU**), stratification (**SDMVSTRA**), and sampling weights (**WTMEC2YR**). By encoding the sampling structure directly into the design object, all subsequent estimation procedures are carried out in a way that is consistent with the NHANES design.

Once the design object is constructed, estimation is performed using survey-weighted functions. The **survey** package applies Taylor series linearization to compute standard errors that reflect the combined effects of clustering, stratification, and unequal weighting. This approach ensures that both point estimates and measures of uncertainty are valid for population-level inference. Using this framework, we estimate population-level quantities such as means, proportions, and regression relationships. These results are presented in the following sections.

6. Finite Population Inference

The following estimates incorporate the NHANES complex survey design and therefore represent design-based inferences for the civilian, noninstitutionalized U.S. population during the 2015–2016 survey cycle. All point estimates, standard errors, and confidence intervals were computed using the survey design object described in the previous section.

6.1 Mean Cholesterol

The estimated population mean for total cholesterol is 185.93 mg/dL, with a 95% confidence interval of 183.45 to 188.41 mg/dL. This estimate reflects the weighted national mean rather than the simple arithmetic average of the observed sample. Because the estimate incorporates the NHANES stratification, clustering, and examination weights, it represents a design-based population quantity rather than a naive sample summary.

6.2 Proportion with High Cholesterol

Using a survey-weighted indicator for elevated cholesterol, we estimate that 10.43% of the target population has high cholesterol, with a 95% confidence interval of 8.97% to 11.88%. Compared with the population means, this proportion provides a more directly interpretable public health measure by describing the share of the population that falls above a clinically meaningful threshold.

6.3 Ratio Estimator

Using *svyratio*, the estimated ratio of total cholesterol to age is **4.45**, with a 95% confidence interval of **4.33 to 4.56**. This result summarizes the average relationship between cholesterol and age at the population level through a ratio estimator, rather than a regression-based effect. Ratio estimators are useful in survey settings because they can improve efficiency when the numerator and denominator are meaningfully related.

6.4 Regression Estimator

To examine the joint relationship between cholesterol and demographic covariates, we fit a survey-weighted linear regression model using age, gender, and race/ethnicity as predictors. The results show that cholesterol increases by approximately 0.63 mg/dL for each additional year of age. Holding age constant, females were estimated to have cholesterol levels about 4.95 mg/dL higher than males on average. Racial differences were less pronounced in the multivariable setting, although the coefficient for non-Hispanic Black respondents was negative and marginally significant. These results suggest that age is the strongest and most consistent predictor of cholesterol in the fitted model, while gender also contributes meaningfully after adjustment.

6.5 Subgroup Analysis

Survey-weighted subgroup estimation reveals clear demographic variation in mean cholesterol levels. Females had a higher estimated mean cholesterol (188.87 mg/dL, SE = 1.43) than males (182.85 mg/dL, SE = 1.83). Across race and ethnicity groups, non-Hispanic White respondents had the highest estimated mean cholesterol (188.47 mg/dL), while non-Hispanic Black respondents had the lowest (178.93 mg/dL). The “Other/Multi-Racial” category produced a relatively large standard error (SE = 4.57), indicating substantially lower precision for that subgroup estimate. This pattern is consistent with the unequal effective sample sizes created by

the complex survey design and highlights why subgroup estimates should be interpreted together with their associated uncertainty.

7. Design Effect & Relative Efficiency

A central objective of this analysis is to evaluate how the NHANES sampling design affects the precision of the estimates presented in Section 6. This is measured using the design effect (DEFF), which quantifies how much the variance of an estimator is inflated due to the complex survey structure relative to a simple random sample (SRS) of the same size.

7.1 Definition

The design effect is defined as the ratio of the design-based variance to the variance under simple random sampling. A DEFF greater than 1 indicates that the complex survey design reduces statistical efficiency by increasing variance.

This loss in efficiency arises primarily from two sources. First, clustering introduces correlation among observations within the same primary sampling unit, reducing the effective sample size. Second, unequal weighting introduces variability across observations, further inflating variance. While stratification can improve precision, it does not fully offset these effects in the NHANES design.

7.2 Empirical Calculation

We evaluate the design effect using the estimated mean total cholesterol from Section 6. Under a naive simple random sampling assumption, the standard error of the mean is 0.49. When the NHANES design is incorporated using the survey package, the design-based standard error increases to 1.27.

This difference demonstrates that the observed sample contains less independent information than an equally sized simple random sample would. Although the nominal sample size is 7,256 individuals, the effective sample size is smaller due to within-cluster similarity and the variability introduced by survey weights.

7.3 Interpretation and Implications

The increase in standard error highlights an important distinction between representativeness and efficiency. The NHANES design, together with survey weights, produces estimates that are nationally representative of the target population. However, this representativeness comes at the cost of increased variance relative to a simple random sample.

If the survey design were ignored, the resulting estimates would appear more precise than they actually are. In this case, relying on the naive standard error would substantially understate uncertainty and could lead to overly confident conclusions about population health metrics. By incorporating the design effect, the analysis provides estimates that more accurately reflect the true variability in the population.

8. Nonresponse Analysis

To produce valid finite population inferences from NHANES, it is necessary to address the role of nonresponse in both the survey design and the analytic sample. Nonresponse arises when selected individuals do not participate in the survey at all or when participating respondents are missing values for specific variables. In NHANES, both forms of nonresponse matter because they can reduce the effective sample size and introduce bias if the missingness is related to health or demographic characteristics. This issue is especially relevant in our study because total cholesterol is measured in the Mobile Examination Center and, therefore, depends on successful completion of the examination component rather than the interview alone.

8.1 Types of Nonresponse

There are two primary forms of nonresponse relevant to this analysis: unit nonresponse and item nonresponse. Unit nonresponse occurs when a sampled individual does not participate in the survey or does not complete the component required for analysis. Item nonresponse occurs when a respondent participates in NHANES but is missing one or more specific variables used in the analysis. In our setting, item nonresponse is particularly important because cholesterol is a laboratory-based outcome that requires a completed blood draw in the Mobile Examination Center. As a result, some respondents may be present in the broader NHANES files but unavailable for this analysis because the relevant laboratory measurement is missing.

8.2 NHANES Adjustment for Nonresponse

NHANES partially addresses nonresponse through its weighting procedure. The survey weights incorporate adjustments that redistribute the representation of nonrespondents to participating individuals with similar characteristics, thereby reducing bias from unequal participation. These adjusted weights are then calibrated to known population benchmarks so that the weighted sample remains aligned with the civilian, noninstitutionalized U.S. population. In our analysis, this adjustment enters through the use of the MEC examination weight, **WTMEC2YR**, which is appropriate for total cholesterol because that variable is measured in the examination component.

At the analytic stage, we handled missing values in the study variables through a complete-case approach. Rather than imputing missing cholesterol or demographic values, we restricted the analysis to respondents with complete data on the variables required for estimation. This restriction was carried out within the survey-design framework so that the retained observations continued to be analyzed with the correct strata, primary sampling units, and weights. In other words, the complete-case sample was not treated as an ordinary subset of observations, but as a restricted survey sample still embedded in the original NHANES design.

8.3 Implications for Inference

This approach has an important implication for interpretation. The weighted estimates remain nationally representative only to the extent that the NHANES nonresponse adjustments and our complete-case restriction adequately preserve the relationship between the observed sample and the target population. Because the weights account for survey-level nonresponse, they help reduce bias from differential participation. However, dropping cases with missing analytic variables may still affect results if the missingness is systematic. As a result, the design-based estimates in this paper should be understood as population inferences based on the complete-case analytic sample after incorporating NHANES nonresponse adjustments through the survey weights.

8.4 Critique and Limitations

A limitation of this approach is that weighting adjustments do not eliminate all possible bias. They are most effective when nonresponse can be accounted for using observed characteristics built into the survey design and weighting process. If missingness is related to unobserved health status or behavioral characteristics, then some bias may remain even after weighting. For example, if individuals with poorer, untreated health were systematically less likely to complete the examination component or provide a usable cholesterol measurement, then the analytic sample could underrepresent that segment of the population. For this reason, the NHANES weighting framework strengthens inference substantially, but it should not be interpreted as a complete correction for every form of nonresponse.

9. Data Processing and Implementation

The analysis was conducted in R using publicly available NHANES 2015–2016 data obtained directly from the CDC. Specifically, the Demographics (**DEMO_I**) and Total Cholesterol (**TCHOL_I**) data files were imported using the **haven** package. These files were selected to

capture both the outcome variable of interest and the survey design variables required for design-based inference.

Relevant variables were then extracted from each dataset. From the demographics file, we retained respondent identifiers (**SEQN**), survey weights (**WTMEC2YR**), primary sampling units (**SDMVPSU**), stratification (**SDMVSTRA**), and demographic covariates, including age, gender, and race/ethnicity. From the cholesterol file, we retained the total cholesterol variable (**LBXTC**).

The two datasets were merged using the respondent identifier **SEQN** to create a unified analytic dataset. Following the merge, variable names were standardized, and the dataset was filtered to include only respondents who completed the Mobile Examination Center (MEC) component of NHANES. This restriction ensures consistency between the analytic sample and the examination weights used in the analysis.

Finally, observations with missing total cholesterol values were removed, resulting in a complete-case analytic sample. This step aligns with the nonresponse handling strategy described in Section 8, where inference is based on respondents with observed values for all variables required in the analysis.

10. Discussion

The results of this analysis demonstrate the importance of accounting for complex survey design when making population-level inferences. Across all estimators, the use of survey weights and design-based methods ensures that the results reflect the structure of the NHANES sampling process rather than the raw observed sample.

One of the most important findings is the difference between naive and design-based measures of uncertainty. As shown in Section 7, incorporating clustering and unequal weighting substantially increases the standard error of the estimated mean cholesterol. This highlights the distinction

between nominal sample size and effective sample size, and reinforces the idea that representativeness does not imply statistical efficiency.

The subgroup and regression analyses further illustrate how demographic factors relate to cholesterol at the population level. Age emerges as the strongest predictor, while gender differences remain meaningful even after adjustment. However, variation in standard errors across subgroups suggests that some estimates are less precise than others, particularly for smaller or more heavily weighted groups.

It is also important to consider the role of nonresponse in shaping these results. While NHANES weights adjust for differential participation, the use of a complete-case analytic sample may still introduce bias if missingness is systematic. As a result, the findings should be interpreted as design-based estimates conditional on both the survey weighting adjustments and the observed data used in the analysis.

11. Conclusion

This analysis applies design-based inference methods to estimate population-level cholesterol measures using NHANES 2015–2016 data. By incorporating survey weights, stratification, and clustering through the Horvitz–Thompson framework, the resulting estimates are representative of the civilian, noninstitutionalized U.S. population.

The findings highlight two key insights. First, properly accounting for the survey design is essential for valid inference, particularly when estimating uncertainty. Second, the NHANES framework allows researchers to translate a relatively small sample into meaningful national estimates, provided that the sampling design and weighting structure are correctly implemented.

At the same time, the analysis underscores the limitations inherent in survey-based inference. Nonresponse, missing data, and unequal weighting all affect the precision and potential bias of

estimates. While the NHANES design mitigates many of these issues, it does not eliminate them entirely.

Overall, this project demonstrates how complex survey data can be used to produce reliable and interpretable population-level estimates when appropriate statistical methods are applied.

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